* Defn $\mathbb{N}$: The set of natural numbers

$$\mathbb{N} := \{1, 2, 3 \ldots\}$$

** Well Ordering Property of $\mathbb{N}$: Every subset of $\mathbb{N}$ has a smallest element

* Defn A set A is called a proper subset of B, ie $A \subsetneq B$, if $A \subset B$ and $A \neq B$.

* Defn Set A and B are called disjoint if $A \cap B = \phi$

* Defn Set theoretic difference of A and B, a.k.a the complement of B relative to A

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$$

* Lemma

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

* Theorem (Principle of Induction) Let $P(n)$ be a that depends upon the natural number n. Suppose that

(1) $P(1)$ is true

(2) If $P(n)$ is true then $P(n+1)$ is also true.

Then $P(n)$ is true $\forall n \in \mathbb{N}$

* Defn The Cartesian product of two set A and B is defined as

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

* Defn A function $f : A \longrightarrow B$ is the subset of $A \times B$ s.t for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x, y) \in f$.

$(x, y) \in f$ is written as $f(x) = y$

* Defn

$$f: \overbrace{A}^{\text{Domain of } f} \longrightarrow \overbrace{B}^{\text{Codomain of } f}$$

Range of f $\quad R(f) := \{ y \in B : \exists x \in A \text{ s.t } f(x) = y \}$

Image / Direct Image of $C \subset A$ $\quad f(C) := \{ f(x) : x \in C \}$

Inverse image of $D \subset B$ $\quad f^{-1}(D) := \{ x \in A : f(x) \in D \}$

* Lemma $\quad f: A \to B$ and $C, D \subset B$ then

(i) $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$

(ii) $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$

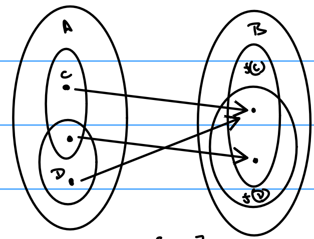
(iii) $f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$

* Lemma $\quad f: A \to B$ and $C, D \subset A$ then

(i) $f(C \cup D) = f(C) \cup f(D)$

(ii) $f(C \cap D) \subset f(C) \cap f(D)$

Proof Hint



Example $\quad f(x) := x^2 \quad C = \{-2\} \quad D = \{2\}$

* Defn $\quad$ A binary relation on a set A is a subset $R \subset A \times A$ which contains the pairs where the relation is valid. Instead of $(a, b) \in R$ we write $\quad aRb$

* Defn $\quad$ Sets A and B have the same cardinality if there exis a bijective map $\quad f: A \to B$

* Defn $\quad$ If a set A has the same cardinality as $\{1, 2, \ldots, n\}$ for some $n \in \mathbb{N}$, we say $|A| := n$.

equivalence class of all set with same cardinality as A $\Big]$ finite cardinality

If A is empty we say $\quad |A| := 0$

* **Defn** $|A| \leq |B|$ If $\exists$ a injective map $f : A \longrightarrow B$

* **Defn** If $|A| = \mathbb{N}$, we say A is countably infinite

* **Defn** If A is finite or countably infinite we say A is countable, else we say A is uncountable

* **Defn** The power set of set A, denoted by $P(A)$, is the set of all subsets of A.

* **Theorem** $|A| < P|A|$, ie there doesn't a surjective map from A to $P(A)$

Proof by contradiction: Let f be a surjective map

$S = \{x \in A : x \notin f(x)\}$, Let $c \in A$ s.t $f(c) = S$

* **Defn** An ordered set S is a set S together with a binary relation '<' s.t

(i) $\forall x,y \in S$, exactly one is true $<, >$ or $=$

(ii) If $x,y,z \in S$ s.t $x>y$ and $y>z$ then $x>z$

* **Defn** Let $E \subset S$ where S is an ordered set

(i) if $\exists\, b \in S$ s.t $\forall x \in E$ $x \le b$ then b is called an upper bound.

(ii) if $\exists\, b_0 \in S$ s.t (a) b_0 is an upper bound

(b) If b is another upper bound then $b_0 \le b$

then b_0 is called the least upper bound or supremum of E

Similary the greatest lower bound / infinum of E is defined.

* **Defn** An ordered set S has the least upper bound porperty if every non empty set of S that is bounded above has a least upper bound.

* **Defn** A set F is called field if it has two operations $(+)$ and $(\times)$ defined on it s.t

① If $x,y \in F$ then $x+y \in F$

② $\forall x,y,z \in F$ $(x+y)+z = x+(y+z)$

③ $\exists\, 0 \in F$ s.t $\forall x \in F$ $x+0 = 0+x = x$

④ $\forall x \in F$ $\exists -x \in F$ s.t $x+(-x) = -x+x = 0$

(① – ④: Abelian group under addition)

⑤ $\forall x,y \in F$ $x+y = y+x$

⑥ If $x,y \in F$ then $x.y \in F$

⑦ $\forall x,y,z \in F$ $(x.y).z = x.(y.z)$

⑧ $\exists\, 1 \in F$ s.t $\forall x \in F$ $1.x = x.1 = x$

⑨ $\forall x \in \{F - \{0\}\}$ $\exists\, x^{-1} \in F$ s.t $x.x^{-1} = x^{-1}.x = 1$

⑩ $\forall x,y \in F$ $x.y = y.x$

(⑥ – ⑩: Abelian group under multiplication)

* **Defn** An ordered field F is an ordered set s.t

(i) If $x,y,z \in F$ s.t $y<z$ then $x+y < x+z$

(ii) If $x>0$ and $y>0$ then $xy>0$

* Lemma Let F be an ordered field and $x, y, z, w \in F$ then

① If $x>0$ and $y>z$ then $xy>xz$

② If $x\neq 0$ then $x^2>0$ Hint $-1\cdot x=-x$

③ $1>0$

④ $0<x<y$ then $x^2<y^2$

⑤ $0<x<y$ then $0<1/y<1/x$

* Lemma An ordered field F has the least upper bound property if and only if it has the greatest lower bound property.

* Defn Real Number, $\mathbb{R}$, is an unique ordered field with least upper bound property s.t $\mathbb{Q}\subset\mathbb{R}$.

* Lemma If $S\subset\mathbb{R}$ is an upper bounded nonempty set, then $\forall\, \epsilon>0\ \exists\ x\in S$ s.t

$$\sup(S)-\epsilon<x\leq \sup(S)$$

* Lemma Let $A, B\subset\mathbb{R}$ be non empty subsets s.t $x\leq y$ $\forall x\in A$ and $y\in B$, then

$$\sup(A)\leq \inf(B)$$

* Theorem If $x, y\in\mathbb{R}$ s.t $x>0$ and $x<y$ then $\exists\, n\in\mathbb{N}$ s.t $nx>y$.

(Archimedian property of $\mathbb{R}$)

Proof Hint: Rephrase $\mathbb{N}\subset\mathbb{R}$ is unbounded above

* Theorem: $\mathbb{Q}$ is dense in $\mathbb{R}$ ie If $x, y\in\mathbb{R}$ s.t $x<y$ then

$$\exists\ r\in\mathbb{Q} \text{ s.t } x<r<y$$

Proof Hint Let $x>0$ $\exists\, n\in\mathbb{N}$ s.t $n(y-x)>1$. $A:=\{k: k\in\mathbb{N} \text{ and } k>nx\}$

$m\in A$ be the smallest element of A. Then $m>nx \Rightarrow x<m/n$

Also $m-1<nx \Rightarrow m<nx+1 \Rightarrow m<ny \Rightarrow m/n<y$

* **Defn** Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$

$$x+A = \{x+y \in \mathbb{R} : y \in A\}$$
$$xA = \{xy \in \mathbb{R} : y \in A\}$$

* **Lemma** If $A \subset \mathbb{R}$ is bounded above, then $\sup(x+A) = x + \sup(A)$

$\inf(x+A) = x + \inf(A)$

If $x > 0$, and A is bounded above $\sup(xA) = x\sup(A)$

$\inf(xA) = x \inf(A)$

* **Defn** Let $A \subset \mathbb{R}$ be a set

(i) if A is empty then $\sup(A) := -\infty$

$\inf(A) := \infty$

(ii) If A is not bounded above then $\sup(A) := \infty$

(iii) " A " " " below then $\inf(A) := -\infty$

* **Defn** When $A \subset \mathbb{R}$ is bounded above s.t. $\sup(A) \in A$ then we denote $\sup(A)$ by $\max(A)$. Similarly $\min(A)$ is defined

* **Defn** Absolute value function $f : \mathbb{R} \longrightarrow \mathbb{R}^+$

$$|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

* **Lemma**

(i) $|x| \geq 0$ and $|x| = 0$ if and only if $x = 0$

(ii) $|xy| = |x|\,|y|$

(iii) $|x| \leq y$ if and only if $-y \leq x \leq y$

(iv) $-|x| \leq x \leq |x| \quad \forall x \in \mathbb{R}$

* Lemma (i) $|x+y| \leq |x|+|y| \quad \forall x, y \in \mathbb{R}$

(ii) $|x-y| \leq |x|+|y|$ "

(iii) $||x|-|y|| \leq |x-y|$ "

* Defn $f: D \to \mathbb{R}$. We say f is bounded above if $\exists M \in \mathbb{R}$ s.t

$$|f(x)| \leq M \quad \forall x \in D$$

* Lemma If $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ are bounded functions s.t

$$f(x) \leq g(x) \quad \forall x \in D \text{ then}$$

$$\sup_{x \in D} f(x) \leq \sup_{x \in D} g(x) \qquad \inf_{x \in D} f(x) \leq \inf_{x \in D} g(x)$$

* Defn Intervals $[a,b] := \{ x \in \mathbb{R} \text{ s.t } a \leq x \leq b \}$

$(a,b) := \{ x \in \mathbb{R} \text{ st } a < x < b \}$

* Lemma All intervals have the same cardinality ie a bijective map b/w any two intervals of $\mathbb{R}$

Example $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \xrightarrow{\tan(x)} \underbrace{(-\infty, \infty)}_{\mathbb{R}}$

* Theorem $\mathbb{R}$ is uncountable

Proof Hint: Let $X \subset \mathbb{R}$ be countably infinite st if $a, b \in \mathbb{R}$ and $a < b$ then $\exists x \in X$ st $a < x < b$

$f: \mathbb{N} \to X$. Let $x_1 := f(1) \quad a_1 := x_1 \quad b_1 := x_{1+1}$

Construct $a_1 < a_2 \cdots < a_n < b_n < b_{n-1} \cdots < b_2 < b_1$ s.t

$x_j \notin (a_k, b_k)$ where $j = 1, 2, \ldots, k$

$A = \{a_1, a_2, \ldots, a_n\} \qquad B = \{b_1, b_2, \ldots, b_n\}$

Let $y = \sup(A)$ then $y \notin A$, $y \notin B$ and $y \notin X$

$\therefore \mathbb{R}$ is uncountable

* **Defn** For an infinite sequence of digits $0.d_1d_2d_3----$

$$D_n := \frac{d_1}{10} + \frac{d_2}{10^2} -- + \frac{d_n}{10^n}$$

* **Lemma** (i) Every infinite sequence of digits $0.d_1d_2--$ represents unique $x \in [0,1]$ s.t

$$D_n \leq x \leq D_n + \frac{1}{10^n} \qquad \forall n \in \mathbb{N}$$

(ii) $\forall x \in (0,1]$ $\exists$ a unique infinite sequence of digits $0.d_1d_2d_3 ---$ that represents x s.t

$$D_n < x \leq D_n + \frac{1}{10^n} \qquad \forall n \in \mathbb{N}$$

* Defn A sequence of real numbers, ie. $\{x_n\}_{n=1}^{\infty}$ is a function $f: \mathbb{N} \to \mathbb{R}$

* Defn $\{x_n\}_{n=1}^{\infty}$ is called bounded if the underlying function is bounded.

* Defn $\{x_n\}_{n=1}^{\infty}$ is said to converge to $x \in \mathbb{R}$ if $\forall \varepsilon > 0 \ \exists M \in \mathbb{N}$ s.t $\forall n \geq M \quad |x_n - x| < \varepsilon$

Limit of the sequence $\lim_{n \to \infty} x_n = x$

* Defn A sequence that is not convergent is called divergent.

* Lemma A convergent sequence has a unique limit.
* Lemma A convergent sequence is bounded.

* Lemma If $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are convergent series s.t $x_n \leq y_n \ \forall n \in \mathbb{N}$ then $\lim_{n \to \infty} x_n \leq \lim_{n \to \infty} y_n$

* Defn $\{x_n\}_{n=1}^{\infty}$ is called monotone increasing if $x_{n+1} \geq x_n \quad \forall n \in \mathbb{N}$
" " " " decreasing " $x_{n+1} \leq x_n$ " "

* Theorem A monotone sequence is bounded if and only it is convergent.
if $\{x_n\}_{n=1}^{\infty}$ is monotone increasing and bounded $\lim_{n \to \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}$
" " " " decreasing " " $\lim_{n \to \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}$

* Lemma For $\{x_n\}_{n=1}^{\infty}$ the following statements are equivalent:
(i) $\{x\}_{n=1}^{\infty}$ is convergent
(ii) The K tail, ie $\{x_{n+K}\}_{n=1}^{\infty}$ converges $\forall K \in \mathbb{N}$
(iii) " " " " " converges for some $K \in \mathbb{N}$

* Def^n Let $\{n_i\}_{i=1}^{\infty}$ be a strictly increasing sequence of natural numbers. $\{x_{n_i}\}_{i=1}^{\infty}$ is called a subsequence of $\{x_n\}_{n=1}^{\infty}$

* Lemma If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence, then all its subsequences, $\{x_{n_i}\}_{i=1}^{\infty}$ are convergent.

* Squeeze lemma: If $\{x_n\}_{n=1}^{\infty}$, $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are sequences s.t
  (i) $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent sequences
  (ii) $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$
  (iii) $a_n \le x_n \le b_n \quad \forall n \in \mathbb{N}$

  then $\{x_n\}_{n=1}^{\infty}$ is convergent and $\lim_{n\to\infty} x_n = \lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$

* Lemma Let $\{x_n\}_{n=1}^{\infty}$, $\{y_n\}_{n=1}^{\infty}$ be convergent sequences then
  (i) $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n + y_n$ converges to $\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$

  (ii) $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n y_n$ converges to $\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n \lim_{n\to\infty} y_n$

Proof Hint: Let $\varepsilon > 0$. $K = \max(|x|, |y|, \varepsilon/3, 1)$ $\quad \exists M_1 \in \mathbb{N}$ s.t $\forall n \ge M_1$ $|x_n - x| < \varepsilon/3K$
$\exists M_2 \in \mathbb{N}$ s.t $\forall n \ge M_2$ $|y_n - y| < \varepsilon/3K$

$\forall n \ge \max(M_1, M_2)$ Prove $|x_n y_n - xy| = |(x_n - x + x)(y_n - y + y) - xy| < \varepsilon$

(iii) if $y_n \ne 0$ $\forall n \in \mathbb{N}$ and $\lim_{n\to\infty} y_n \ne 0$ Then $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n / y_n$ converges to $\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n / \lim_{n\to\infty} y_n$

Proof Hint: Prove $\{1/y_n\}_{n=1}^{\infty}$ converges to $1/y$ where $y = \lim_{n\to\infty} y_n$

Let $\varepsilon > 0$ $\quad \exists M_1 \in \mathbb{N}$ st $|y_n - y| < \min\left(\frac{|y|^2 \varepsilon}{2}, \frac{|y|}{2}\right)$

$\forall n \ge M_1$ $\quad |y| = |y - y_n + y_n| \le |y - y_n| + |y_n| < \frac{|y|}{2} + |y_n| \Rightarrow \frac{1}{|y_n|} < \frac{2}{|y|}$

$$\left|\frac{1}{y_n} - \frac{1}{y}\right| = \left|\frac{y - y_n}{y y_n}\right| = \frac{|y - y_n|}{|y|\,|y_n|} < \frac{|y|^2 \varepsilon/2}{|y|} \times \frac{2}{|y|} < \varepsilon$$

* **Lemma** If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence s.t $x_n \geq 0$ $\forall n \in \mathbb{N}$, then

(i) $$\lim_{n\to\infty} \sqrt{x_n} = \sqrt{\lim_{n\to\infty} x_n}$$

(ii) $$\lim_{n\to\infty} |x_n| = |\lim_{n\to\infty} x_n|$$

* **Lemma** Let $\{x_n\}_{n=1}^{\infty}$ be a sequence, $x \in \mathbb{R}$ and $\{a_n\}_{n=1}^{\infty}$ be another sequence s.t

(i) $\{a_n\}_{n=1}^{\infty}$ is convergent

(ii) $\lim_{n\to\infty} a_n = 0$

(iii) $|x_n - x| < a_n$ $\forall n \in \mathbb{N}$

then $\{x_n\}_{n=1}^{\infty}$ is convergent and $\lim_{n\to\infty} x_n = x$

* **Lemma** Let $c > 0$

(i) If $c < 1$, the sequence $\{c^n\}_{n=1}^{\infty}$ Converges to 0

(ii) If $c > 1$, " " " is unbounded

* **Lemma** Let $\{x_n\}_{n=1}^{\infty}$ be a sequence s.t

(i) $x_n \neq 0$ $\forall n \in \mathbb{N}$

(ii) $L = \lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|}$ exists

then (a) if $L < 1$ then the sequence $\{x_n\}_{n=1}^{\infty}$ converges to 0

(b) " $L > 1$ " " " " is unbounded

* **Defn** :- Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Define $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ sequences where $a_n := \sup\{x_k : k \geq n\}$ $b_n := \inf\{x_k : k \geq n\}$

Limit superior $\limsup_{n\to\infty} x_n := \lim_{n\to\infty} a_n$

Limit inferior $\liminf_{n\to\infty} x_n = \lim_{n\to\infty} b_n$

* **Lemma** For a bounded sequence $\{x_n\}_{n=1}^{\infty}$

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} a_n = \inf\{a_n : n\in\mathbb{N}\}$$

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} b_n = \sup\{b_n : n\in\mathbb{N}\}$$

Also $\liminf\limits_{n\to\infty} x_n \le \limsup\limits_{n\to\infty} x_n$

* **Theorem** If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence, then $\exists$ subsequences $\underbrace{\{x_{n_i}\}_{i=1}^{\infty}, \{x_{m_j}\}_{j=1}^{\infty}}_{\text{Need not to be monotone}}$ s.t

$$\limsup_{n\to\infty} x_n = \lim_{i\to\infty} x_{n_i}$$

$$\liminf_{n\to\infty} x_n = \lim_{j\to\infty} x_{m_j}$$

~~Proof~~ Hint: Let $n_1 = 1$. Let $n_1, n_2, \ldots, n_{k-1}$ are already defined

$m\in\mathbb{N}$ s.t m s.t $\forall n \geq m$ $|a_{n_{k-1}+1} - x_n| < 1/k$. Define $n_k = m$

then $\forall k \geq 2$ $|a_{n_k} - x_{n_k}| < \frac{1}{k}$ as $a_{n_k} \le a_{n_{k-1}}$

$\{a_{n_k}\}_{k=1}^{\infty}$ is subsequence of $\{a_n\}$ $\Rightarrow$ $\lim\limits_{n\to\infty} a_n = \lim\limits_{k\to\infty} a_{n_k} = a$

Let $\varepsilon > 0$, $\exists M_1 \in\mathbb{N}$ s.t $\forall i \geq M_1$ $|a_{n_i} - a| < \varepsilon/2$

$M_2\in\mathbb{N}$ s.t $1/M_2 < \varepsilon/2$ $\qquad J = \max(M_1, M_2, 2)$

$\forall i \geq J$ $\quad |x_{n_i} - a| \le |x_{n_i} - a_{n_i}| + |a - a_{n_i}| < \frac{1}{i} + \varepsilon/2 < \frac{1}{J} + \varepsilon/2 \le \frac{1}{M_2} + \varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$

* **Theorem** Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Then $\{x_n\}_{n=1}^{\infty}$ converges if and only if

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n$$

Also $\liminf\limits_{n\to\infty} x_n = \limsup\limits_{n\to\infty} x_n = \lim\limits_{n\to\infty} x_n$

* **Lemma** If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence and $\{x_{n_k}\}_{k=1}^{\infty}$ is a subsequence, then

$$\liminf_{n\to\infty} x_n \le \liminf_{k\to\infty} x_{n_k} \le \limsup_{k\to\infty} x_{n_k} \le \limsup_{n\to\infty} x_n$$

* Lemma: A bounded sequence $\{x_n\}_{n=1}^{\infty}$ is convergent to x if and only if every convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ converges to x

* Theorem: Bolzano-Weierstrass: Suppose a sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers is bounded. Then there exists a convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$

* Defn: $\{x_n\}_{n=1}^{\infty}$ diverges to infinity, i.e. $\lim_{n\to\infty} x_n = \infty$ if $\forall K \in \mathbb{R}\ \exists M \in \mathbb{N}$ s.t

$$x_n > K \quad \forall \ n \geq M$$

$\{x_n\}_{n=1}^{\infty}$ diverges to $-\infty$, i.e. $\lim_{n\to\infty} x_n = -\infty$ if $\forall K \in \mathbb{R}\ \exists M \in \mathbb{N}$ s.t

$$x_n < K \quad \forall \ n \geq M$$

* Defn: Extension of defn of lim sup and lim inf to unbounded sequences

* Defn: A sequence $\{x_n\}_{n=1}^{\infty}$ is called a Cauchy sequence. If $\forall \epsilon > 0, \exists$ an $M \in \mathbb{N}$ s.t

$$\forall \ k \geq M, \ m \geq M$$

$$|x_m - x_k| < \epsilon$$

* Lemma: If a sequence is Cauchy, it is bounded

* Theorem: A sequence of real numbers is Cauchy if and only if it is convergent.

Proof Hint: Cauchy $\Rightarrow$ bounded $\Rightarrow \exists \ \{x_{n_i}\}_{i=1}^{\infty}$ s.t $\lim_{i\to\infty} x_{n_i} = \limsup_{n\to\infty} x_n = a$

$\hookrightarrow \exists \ \{x_{m_j}\}_{j=1}^{\infty}$ s.t $\lim_{i\to\infty} x_{m_i} = \liminf_{n\to\infty} x_n = b$

$\exists M_1 \in \mathbb{N}$ s.t $\quad \forall i \geq M_1 \quad |x_{n_i} - a| < \epsilon/3$

" $M_2 \in \mathbb{N}$ " $\quad \forall j \geq M_2 \quad |x_{m_j} - b| < \epsilon/3$

" $M_3 \in \mathbb{N}$ " $\quad \forall i \geq M_3 \quad |x_{n_i} - x_{m_j}| < \epsilon/3$

Let $i = \text{Max}(M_1, M_2, M_3)$

$$|a-b| = |a - x_{n_i} + x_{n_i} - x_{m_i} + x_{m_i} - b|$$

$$\leq |x_{n_i} - a| + |x_{m_i} - b| + |x_{n_i} - x_{m_i}|$$

$$< \epsilon/3 + \epsilon/3 + \epsilon/3 < \epsilon$$

$$\therefore a = b$$

* **Defn** Given a sequence $\{x_n\}_{n=1}^{\infty}$, a series is defined as

$$\sum_{n=1}^{\infty} x_n .$$

A series is said to converge if the sequence of partial sums $\{s_k\}$ converges where $s_k := \sum_{n=1}^{k} x_n = x_1 + x_2 + \dots + x_k$

* **Lemma** Let $\sum_{n=1}^{\infty} x_n$ be a series. Then $\sum_{n=1}^{\infty} x_n$ converges if and only if $\sum_{n=M}^{\infty} x_n$ converges $\forall M \in \mathbb{N}$

* **Defn** A series $\sum_{n=1}^{\infty} x_n$ is called Cauchy if the sequence of partial sums $\{s_k\}$ is Cauchy.

* **Lemma** The series $\sum_{n=1}^{\infty} x_n$ is Cauchy if and only if $\forall \varepsilon > 0$ $\exists$ an $M \in \mathbb{N}$ s.t $\forall n \geq M$ and every $k \geq n$

$$\left|\sum_{i=n+1}^{k} x_i\right| < \varepsilon$$